

Lec 40 Cycle GAN

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Transcript

[00:00] Hello everyone. Welcome back to the lecture series on fundamentals of generative AI and large language models. In today's session, we will discuss about cycle consistent adversarial networks, which we call as CycleGANs. So, the contents for today's session goes like this. What is the motivation for CycleGAN? Introduction to CycleGAN, architecture of CycleGAN, cycle consistency, loss functions, in that one we have adversarial loss and cycle consistency loss, and objective function.

[00:37] So, we will start with what is the motivation for the discovery of CycleGAN. So, we have seen in traditional GANs or in pix2pix, which is required for image-to-image translation, we required paired data set. Like, suppose I have an input image and a target image, like I have a sketch of a bag uh as the input image, and the target image I want a photo of that bag or something like that. Uh you know, daytime image, I want to convert it into a nighttime image or a painting or a picture, I want to change it to some particular style of painting. So, in all these cases, in all these image-to-image translation work, we have seen we required the paired data set. So, I want both the input image as well as the

[01:35] target image for my model to work properly. But, collecting such paired data set is very difficult. It is highly possible that we may not have the sketch of the same

bag and the the photo of the same bag. Sketch of the bag and the photo of the same bag. Okay, so it is very much possible that paired data sets are not available. Okay? So, the question that came to the mind of scientists was can we have a model that can learn the translations between two domains usually using only separate image collections without paired examples. So, if I have the sketch of a bag without having the photo of the bag with me or some autumn picture without having the winter you know, winter condition picture can I use these single pictures for a

[02:38] translation from the input image to the target image, input domain to the target domain. Is it possible? So, that was the question that came to the mind of scientists and from there the motivation came for CycleGAN. So, we will look into what exactly the CycleGAN is, how it works, what is its architecture and all those things. Okay? So, as I was telling, CycleGAN learns image-to-image translation. It is used for image-to-image translation between two domains like a sketch and a photo without requiring paired examples. Without paired examples itself, CycleGAN can produce one photo of a bag from the sketch of the bag.

[03:28] Here, we have two photos. One is the photo of a horse, another is the photo of a zebra. So, from horse, we can translate to zebra using CycleGAN and once again, from zebra, we can go back to horse. So, this is this is the thing which is possible in CycleGAN. Similarly, a photo of a place, how it will look in winter and how it will look in summer. So, that forward translation, winter to summer, as well as summer to winter, is possible in CycleGAN. Another example is a normal photo of a flower, then we have changed it to some painting style, and vice versa is also possible using the CycleGAN.

[04:16] We call these photos the actual input one, which we are giving to the model, we call it as the source domain, and what we want it to the trans- what translation we want from that source image, we call it as the target domain, okay? The source domain and target domain we will be using quite often throughout my lecture, okay? So, this concept of CycleGAN was first introduced by Zhu Jun-Yan Hu and his team in the paper Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks.

[04:56] This scientists belong to AI research lab, UC Berkeley. So, they are the people who have come up with this architecture, and it is widely widely used now. Now, we will try to see what are the components of CycleGAN architecture. So, we know right now the problem statement that we have in hand, or what is the thing that we will go we are discussing, will be a transition of or translation of a horse's image to a zebra's image, okay?

[05:32] So, this is my domain X, horse's image is my domain X, and zebra is my domain Y. So, we have two separate domains, one domain is X, another domain is Y. My X have the horses image, my domain Y have the zebras image, okay? Now, as

there are two domains here, we have two generators in case of CycleGAN. So, we will look into what are these generators, what are their work, and how they work. So, there are two generators that are as I was telling, generator G and generator F.

[06:13] What is the function of generator G? Generator G takes input as X and converts it into Y. Generator F takes Y as input and converts it to X. So, this conversion X to Y is done by G and this conversion Y to X is done by generator F. Okay? So, similarly, we have two discriminators, DY and DX. Now, we have to remember all throughout this lecture we will remember my domain Y is basically zebra. And my domain X is horse, okay? So, what is the function of DY?

[07:04] DY checks whether the zebras images are real or generated, okay? And what will be the function of DX in the similar way? It will check whether the horses images are real or generated. Okay? So, in this architecture framework, we have two generators, G and F. G takes from or translates the image from X domain to Y domain. F generator translates the image from Y domain to X domain, and there are discriminators dy and dx. dy checks whether the images of the zebras are real or generated. dx checks whether the horses images are real or generated.

[07:56] Together with that one, we have a very very important concept which is called as the cycle consistency mechanism. We will look into what exactly it is in a in one or two slides after this, okay? So, this is the key idea introduced in CycleGAN. And it plays a very very significant role in the working of the CycleGAN architecture. Now, we will have a look into what are the generators that are used and how they work. So, as I was telling in my previous slide that we have two generators g and f and these two generators, they are trained.

[08:36] g transforms images from domain x to domain y. Domain x is horse and y is zebra. And f transforms from domain y to domain x. This I have already told, but reinforcement is always always good. I have a picture here. This picture I will be using in couple of my next slides also. Just just to have this picture in front of our eyes so that we don't get confused. So, g takes from x to y. y f takes you from y to x. And dx is the discriminator here.

[09:15] Which checks for the real or generated images of the horses images. dy checks for whether zebras images are real or generated, okay? So, once again, I have put it here. The conversion can be both ways. So, this goes for g and this is for f. And once again, I have written G is horse to zebra, zebra to horse. And the generator mapping functions are G is X to Y and Y to X is my F. Okay? Now, coming to the discriminators, there are two adversarial discriminators DY and DX.

[10:00] Once again, so DY is connected with the domain Y, DX is connected with the domain X. And once again, just to make it, you know, see sink inside, I have once again, I have put horse is your X domain, zebra is your Y domain, and DY, what is the function of DY? It distinguishes real images from domain Y and the generated images

GX. Now, what do I mean by this GX? When X is undergoing a transformation using the generator G, we call it as GX.

[10:40] Okay? Whereas DX distinguishes real images from domain X and the generated images FY. What is this FY? If you take input image as Y, and then you pass it through the generator F, whatever generated images you are getting, that is your FY. Okay? So, input image as Y going through the generator F, the output image will be X. Okay? So, that is this transition. So, from zebra to horse. So, these are the four networks, two generator networks, and two discriminator networks that are there and they are learned during the training phase.

[11:31] Once again, DX checks whether the horse's image is real or fake. DY checks whether the zebra's image is real or fake. Okay. Now, we are coming to the very, very important concept which is called as the cycle consistency in CycleGAN. What is that cycle consistency? We have domain X which is my horse. G is the generator which is taking from horse to zebra, making translating horse's image to zebra's image and you get a zebra's image and then again, if you pass this image from through F then the zebra should go back to you know, translate back to horse. Okay.

[12:20] So this will be my final after that translation is done through F, generation is done through F we get this horse's image. So, basically, what is happening here is from horse's image through generator G I am getting zebra's image. Now, that zebra's image, I am passing through the generator F and again getting back horse's image. Now, what does this cycle consistency mean? Or what it tries to enforce. Cycle consistency tries to enforce that the horse that is generated using the generator F should be same as the original one, original horse.

[13:09] Otherwise, what happens? Any horse will go and change into any zebra and when I am trying to reconstruct it back to horse that horse may be different from the original horse. So, the cycle consistency is a mechanism which enforces that when I translate a picture from horse to zebra and again from zebra to horse this this final horse should be same as my original horse. Okay? Similarly when we go back in the backward direction, that means from zebra to horse, we I have the zebra's image here.

[13:49] Zebra to horse translation is done using F generator, if you remember. And then we get a horse's image. And then once again if we pass it through generator G I get the zebra's image and cycle consistency enforces that this zebra should be same as the original zebra. Okay? So, this is the idea of cycle consistency in CycleGAN. And this is a very very important concept in CycleGAN. What exactly we are enforcing here in cycle consistency, as I told, once again I have written it here also.

[14:31] The translated images should preserve the underlying contents or structure of the original image. So, that it should not deviate from the original images. Whatever be the number of translations, the original underlying structure or content

should be preserved. Okay? So, that is the main idea of cycle consistency. So, we have understood till now the architecture of CycleGAN. It has two generators, two discriminators. And then we have seen what is the cycle consistency. Now we have come to loss function in CycleGAN.

[15:07] What is the loss function? So, as I know, there are four networks, two generator networks and two discriminator networks, right? And if I have to you know, number them, The first one is the generator G, which takes from horse to zebra. Then we have the discriminator DY, which checks the zebra's images that are that we have, are they real or generated? These are two uh networks. Then we have the third network F, which is a generator network, which translates zebra to horse.

[15:52] And then we have the discriminant network, which is DX. If it checks if the images of the horses are real or not. Okay? So, the four networks are there, and the loss function in CycleGAN have to look into all the losses that are coming from all these networks, okay? Now, in CycleGAN, there are two losses. One is called as the adversarial loss, another one is called as the cycle consistency loss. Okay? What is the function of adversarial loss? Adversarial loss checks for the checks in the image whether it is real or generated. Whereas cycle consistency loss looks into whether that cycle consistency is maintained in the model, okay?

[16:46] So, in adversarial loss, as there are two generators and two discriminators, there are two adversarial losses. So, we have to see what are those two adversarial losses. One adversarial losses uh one of the adversarial loss is having G and DY. Another one is with F and DX, okay? So, these are the two adversarial losses. So, in the loss function, we have two loss coming in picture. One is the adversarial loss, one is the cycle consistency loss. Adversarial loss checks for whether the images, whether it is the zebra's image or the horses' image, they are real or generated. And cycle consistency lock loss will check into whether that cycle consistency, that means horse to zebra to horse, whether these two horses

[17:34] are same or not. Okay? So, that is the function of cycle consistency loss. Now, we are looking into adversarial losses. So, there are two losses, G and DY, and another one with F and DX. Okay? So, we will look into the loss that is coming from G and DY. G is the generator, which takes from horse to zebra. DY is the discriminator, which checks if my zebra's images are real or generated. Okay? So, the expression for the loss, this adversarial loss looks like expectation value over all Y they belong to the probability they belong to the distribution data $Y \log DY$ of Y plus expectation value over all X, which belongs to the X data set, $\log 1 - DY GX$. Okay? So, this is the expression.

[18:41] Now, we will come into understanding what each and every term means. Okay? Now, we will see what is the first term giving. The first term term is training the discriminator DY to classify real zebra images as real. So, we all know DY, what

is D_Y ? Whether the zebra image is real or fake, okay, or generated. Okay? That is what is the function of d_y ? Now, the first term in the loss function, it ensures that the discriminator gets trained properly.

[19:23] Discriminator d_y gets trained properly to classify real zebra images or distinguish real zebra images from generated images, okay? Whereas, the second term, it enforces that the discriminator d_y should be trained to classify generated zebra images as fake. So, the first term ensures that the discriminator classifies real zebra images as real. The second term ensures the discriminator d_y classifies generated zebra images as fake.

[20:06] Okay. So, these are the two terms of the adversarial loss that is coming from g and d_y . What is the desired output and what exactly we want to minimize here? That we have to see. So, the desired output is d_y of fake zebra or the generated zebra should be should tend to zero. Correct? Because discriminator d_y will check if the zebra's image is real or generated. If it is real, d_y should give out a score, probability score, which is near to one.

[20:44] If it is a generated image or a fake image, then it should give a score which tends to zero, okay? Now, the function of the generator, as we all know in GANs, the function of the generator is to fool the discriminator. So, what the generator would like to do? Generator would want when d_y is looking into the images generated by g , it gives a probability score of one. That means the generate the discriminator should get fooled looking into the generated images. It should not be able to distinguish it from the original images.

[21:31] Okay? So, what is this g_x ? g_x is basically your generated zebra's image. Okay? When the generator is g . So, the generator tries to minimize this loss and discriminator d_y tries to maximize this loss. Okay? Because discriminator will always want to give high score for the real images and scores very close to zero for generated images, but the opposite is the desire of the generator. Okay? Then we will come to the second adversarial loss that is coming from f and d_x .

[22:15] So, we all know once again I have kept just to keep it in you know in our in front of our eyes because there are four networks, g , f , d_x , d_y . So, I have kept in all the slides just to reinforce this one that you know, f is the generator which takes zebra to horse, translates zebra's image to horse's image, and d_x is the discriminator which checks if the images of the horses are real or not. Okay? And this is the figure for that one. So, f is this uh generator and d_x is the discriminator.

[22:54] So, we will look into the mathematical equation here. It looks like this. It has two terms. One is with $\log d_x$ of x and the expectation value is taken over all the x x belongs to the uh you know distribution of uh data set x . And the second term is the distribution expectation value of all the data uh the uh expectation value of you know $\log$ of $1 - d_x$ of f of y and the expectation is taken over all the uh uh points which

are coming from the distribution u data distribution y .

[23:31] Now we will once again we will see what are these two terms trying to do. So the first term trains the discriminator D_x to classify real horses images as real just exactly as the last one. And the second term trains the discriminator D_x to classify generated horse images as generated or fake, okay? Generated it should be different from the real images. And the desired output in this case will be D_x of fake images should be tending to zero fake images or the generated image we can tell it as generated images. And the generator f will try to fool the discriminator D_x . That's the main idea.

[24:11] And f would want $D_x f(y)$ tending to one. What exactly it means? What is $f(y)$? f So when f function or the f generator is working on any y that means f this y is your zebra image the zebra image and then when it is going through the u generator f what it is supposed to give you this is your zebra it is supposed to give you a horse's image. But this horse's image is basically a generated horse's image. It is not a real horse's image.

[24:51] So what the generator would like is they want the $f(y)$ when it is going through the discriminators check which is the discriminator working here which is a D_x . It should give out a probability score tending towards one near to one. That means the generator f is trying to fool the discriminator D_x . Okay? Now as I was telling before, adversarial losses only ensures output looks realistic. It is not ensuring that the output corresponds to the input.

[25:28] Like I was telling in cycle consistency, if I translate from horse to zebra and again from that zebra back to the horse, those two horses, my original horse and the generated horse, those two images should be same. Okay? But that is not enforced by adversarial loss. So we have something called as the cycle consistency loss in cycle gan. So what is the cycle consistency losses equation? So it There are two terms here.

[25:58] One is the you know u This what is this one? This is the difference between the generated horses image and the actual horses image. We have to remember my x domain is horses domain. What is this $f(gx)$ means? $f(gx)$ means suppose so my x when it goes through g I get gx . Which is x is my horse. gx is my zebra. And then once again when it goes through f $f(gx)$ What is this gx ? gx is zebra. That means it gives me f of zebra which gives me again the horse.

[26:52] Okay? So, this horse is my $f(gx)$. And this horse is my x . So, it takes the absolute difference between it is the L1 norm of the generated horse's image and the actual horse's image. And the expectation value is taken over all the x 's in the distribution data x , okay? And here also the expectation value is taken. For this case, it is the difference or the mean absolute error between the actual zebra's image and the zebra's image that has been generated, okay? So, what exactly it is doing? What

is my g of f of y ?

[27:47] If I take again once again, if I take my input as y , which is my zebra, then when I send it through f , I get f of y , which is nothing but my horse. Again, I pass it through my generator g , so it gives me g of f of y , which is what? It should be another zebra. So, this is my this zebra. And the this one y is my original zebra. So, the difference between these two is taken. The first term is called as the forward cycle loss.

[28:31] It compares between the original horse versus the reconstructed horse. Whereas, the second term is the backward cycle loss, which you you know, compares between the original zebra versus the reconstructed zebra, okay? And we have to minimize or the model have to minimize both these errors. Okay. So, that's the main idea. So, whatever I have shown here in writing, the same thing I have shown here, so that it becomes easier.

[28:59] And then once this cycle consistency losses we have got we have got. So, by this time we have got adversarial losses and cycle consistency loss. So, now we are in a shape where we can talk about the objective function of CycleGAN. So, what is the objective cycle objective function? For that one we'll just have a look into the complete CycleGAN loss. So, CycleGAN loss the total cycle CycleGAN loss is this one. It have both the adversarial components, one coming from G and D_Y , another coming from F and D_X , and then we have the cycle consistency loss multiplied by a factor λ . Okay. So, that is my complete CycleGAN loss.

[29:42] So, as I was saying, this is the adversarial loss for horse to zebra transition, and this is for adversarial loss for zebra to horse translation. Okay. And this is the cycle consistency loss. In the original paper, they have taken λ equal to 10. Okay. So, that is the weightage that is given for the uh cycle consistency loss. Now, the objective function can be written as $\arg \min$ of G and F because generator always wants to minimize the loss, and discriminate does tries to maximize the loss, and then we have the loss function. So, this is the total loss function of CycleGAN, $\arg \min G$ and F , $\max$ of D_X and D_Y , and then we have the total loss here. So, that is the objective function of CycleGAN.

[30:34] So, the idea is to find the values of G and F that minimizes the loss because a generator always any generator G and F , always tries to fool the discriminator, maintain cycle consistency, uh that is minimize the cycle consistency loss. Whereas for D_X and D_Y , they always tries to maximize the uh loss and uh because they want to classify real and fake you know, generated images correctly and by $\max$ and that is done by maximizing the loss.

[31:04] Okay, so this is about the objective function of CycleGAN. These are some of the transformations taken from the original paper. So you can see the input is horse and output is zebra and there are various kind of horses and zebra that is coming.

And uh then you have uh you know, this is zebra to horse this this is completely zebra to horse transitions for various shapes and poses and other things. Then winter to summer.

[31:33] As I was telling in the beginning, then summer to winter transition, orange to apple and this photos to different styles of painting. Okay, Van Gogh and other styles of painting. So these images are taken from the original paper and this is and because of the cycle consistency and uh the other adversarial losses or the architecture of the entire um framework, these transformations uh work very good and they maintain the original and the generated images, they look both realistic as well as keeping the uh um properties or the structure preserved even though through the multiple rounds of transitions. Okay.

[32:20] So with this one I come to the end of my session today. So before I conclude, just have a quick look. So we have seen uh what is the motivation for this CycleGAN. Uh the motivation was um in Pix2Pix we have seen that we required paired image data sets for trans image to image transition uh translation. Um But getting you know, paired data sets are difficult and it may be expensive. So, there was a question that came to the mind of the scientists that can we do this kind of image to image translation with without the paired image data set.

[32:58] That is the main motivation for coming into cycle again. And in cycle again, we have seen how uh you know, from one image of one domain can be changed to the other domain. What are the components in that network? There are two unlike our normal GANs here in this case, there are two generators, two discriminators taking care of the two different domains X and Y. And we have seen how these transition occurs. What is the function of each and every generator? What is the function of each and every discriminator? And then we have seen what are the losses that are What is cycle consistency mechanism?

[33:38] Cycle consistency mechanism is basically when you change a horse's image to a zebra and then back to a horse's image, the final image of the horse should be same as the original horse's image. So, this is the concept of cycle consistency which is very very important for cycle GANs working. Otherwise, any horse will go to any zebra. There will be no matching between the reconstructed horse and the original one. Okay. So, then after understanding the cycle consistency loss, we cycle consistency mechanism, we have come to the loss functions that are working in the cycle GAN. So, we have seen adversarial losses which takes care of the uh what takes care of whether the image that is generated looks realistic or

[34:25] not. And then there is a cycle consistency loss which looks into the matching of the original image with the reconstructed image, okay? And then we have seen in cycle adversarial losses also there are two terms coming from one discriminator and one generator that is one set, another one is the second generator and the second

discriminator. So there are two loss terms that is coming in the adversarial loss and then we have the cycle consistency losses. We have seen the mathematical expression and the total loss of the cycle GAN basically is um addition of all these losses with a factor of lambda multiplied with the cycle consistency losses. Cycle consistency loss and this value of lambda has been taken as 10 in the

[35:11] original paper. So that is the weightage that has been given for the cycle consistency loss. And once we have got the total complete loss function, we have looked into the objective function and what is the objective of that entire model? The generators try to fool the discriminators by minimizing the losses. The discriminators try to you know, push the generator to make better images or they try to classify real and generated images correctly by maximizing the losses. So that's about the working of cycle GAN.

[35:49] And in my next session, I will be discussing about style GAN that is another variation of GAN and till that time, I urge all the students or participants to go through the original paper and look into the mathematical things more in details. Those are the things that are given more in details there, mathematical steps. And uh until we meet for the next session, thank you all and happy learning.